

Effects of Menstrual and Reproductive Factors on the Risk of Breast Cancer: Meta-analysis of the Case-Control Studies in Japan

Chisato Nagata, Yao-Hua Hu and Hiroyuki Shimizu

Department of Public Health, Gifu University School of Medicine, 40 Tsukasa-machi, Gifu 500

To elucidate the magnitude of the effect of menstrual and reproductive factors on breast cancer occurrence among Japanese women, we reviewed eight case-control studies previously conducted in Japan and used a quantitative method (meta-analysis) to summarize the data. While individual studies have different methods and populations, the estimated odds ratios (ORs) in the studies were statistically homogeneous for all study variables. It was confirmed that early age at menarche, late age at first birth, and premenopausal status are significantly associated with risk of breast cancer; an estimated combined OR of 0.68 (95% confidence interval (CI): 0.59–0.77) was obtained for women with onset of menstruation after age 16 compared to those before age 14. Nulliparous women had higher risk than women with first birth before age 25 (OR = 1.56 95% CI: 1.27–1.91). The OR for women with first birth after age 35 was 2.26 (95% CI: 1.85–2.77) compared to women at first birth before age 25. Premenopausal women had a higher risk than women with menopause before age 50 (OR = 2.21, 95% CI: 1.53–3.20). We also found a significant protective effect of high parity after controlling for age at first birth and the other menstrual factors. The OR estimate for 3 or more births compared to nulliparity was 0.68 (95% CI: 0.54–0.86). The meta-analysis provided quantitative estimates of breast cancer risk among Japanese women with improved precision.

Key words: Breast cancer — Case-control study — Japan — Meta-analysis — Risk factor

Numerous epidemiological studies have been devoted to finding the risk factors of female breast cancer. Timings and occurrences of menstrual and reproductive events, i.e., age at menarche, first birth, and menopause, are well known, established risk factors of breast cancer.¹⁾ A protective effect of increasing number of births has also been demonstrated in many previous studies.^{2–6)}

The recent increase in breast cancer incidence in Japan is noteworthy. The age-adjusted incidence rate of breast cancer in Japanese women of Miyagi was 27.8 per 100,000 in 1983–1987, about 2.3 times the comparable rate in 1959–1960.⁷⁾ This increase might reflect changing prevalences of risk factors of breast cancer among Japanese women. We are interested in establishing to what extent the change in the distributions of major determinants of breast cancer can explain the recent increase in breast cancer incidence in Japan. To resolve this problem, we need to estimate precisely the effects of menstrual and reproductive factors on breast cancer in Japanese women. We undertook an appraisal of analytic epidemiological studies on breast cancer which have been conducted in Japan and applied the methods of meta-analysis to summarize the outcome of the studies and combine them. This method provides increased statistical power to improve estimates of effect size and inferences about conflicting conclusions among the component studies.

MATERIALS AND METHODS

The epidemiological studies of breast cancer among Japanese women with reference to menstrual and reproductive events were identified by literature search on Medline from January 1966 to March 1995 and by checking the reference lists of reports already retrieved. From 14 studies identified,^{8–20)} 7 were excluded from the present meta-analysis because the results published before were included in the review articles,^{15, 16)} information necessary for the present analyses was lacking,¹⁷⁾ the results were based on Japanese A-bomb survivors,¹⁸⁾ or the studies were conducted repeatedly using patients of the same hospital at different time.^{19–21)} In the last case, only the most recently published results were included in this analysis. We assessed the study quality of the remaining 7 and an additional one eligible unpublished report (Hu *et al.*, submitted) using a checklist and decided to include all of them in the present meta-analysis. The checklist included criteria referring to predetermined method, unbiased data collection, avoidance of bias in selection of cases and controls, equal diagnostic examination and surveillance, and avoidance of the effects of prognostic factors.^{22, 23)}

All of the studies were case-control studies and we chose the odds ratio (OR) as the measure of effect. We analyzed the data in the original studies to extract study-specific OR estimates and their standard error (SE)

estimates.²⁴⁾ If confidence interval was given instead of SE, we calculated the SE as $(\log(OR_u) - \log(OR_l))/3.92$, where OR_u and OR_l represent the upper and lower confidence limits of the OR. When only the distribution of cases and controls with respect to exposure is available, the OR and SE are approximated as ad/bc and $1/a + 1/b + 1/c + 1/d$, respectively (a , b , c , and d indicate numbers of exposed cases, exposed controls, unexposed cases, and unexposed controls, respectively). Exposures were treated in a categorical fashion in all reports. We set up the standard categories for exposure variables and converted the original estimates so as to fit the new categories by combining the category-specific ORs or using weighted least-squares linear regression models.^{24, 25)} For the latter, category midpoints were assigned to the categories of exposures defined by the original authors. For open-ended categories, the midpoints were estimated based on the distributions of exposures among the other study subjects or participants in the national survey of marriage and fertility in Japan.²⁶⁾ The method proposed by Greenland²⁴⁾ was utilized to estimate the adjusted OR for age and other menstrual or reproductive variables.

After these preliminary steps, we combined the data across the studies. The weight given to each $\log(OR)$ is its variance from the individual studies.²⁷⁾ The homo-

geneity assumption was tested by using the chi-square test.²⁷⁾ This assumption states that the studies should on average yield the same value, so that differences between the estimates are entirely due to random error.²⁴⁾ All statistical analyses were performed using PC-SAS.²⁸⁾

RESULTS

Table I gives details of the 8 studies that were analyzed in the present meta-analysis. Five of them included hospital controls. In the earliest study, the authors had selected controls available at public health centers. Neighborhood controls were used in the remaining 2 studies, but controls were selected from the general population in one and limited to the participants in breast cancer screening in the other. All but 3 studies utilized a self-administered questionnaire for data collection. Although some potential sources of bias or methodological weakness (e.g., selection bias, Berkson's bias, recall bias, or insufficient study power) were indicated, histologically confirmed incident cases and controls were selected in a random fashion in all studies. The data were collected in a similar setting for cases and controls in every study with hospital controls. Relatively high response rates (at least 69%) were obtained even in the studies with neigh-

Table I. Case-Control Studies Included in the Meta-analysis

Study	Place	Time cases diagnosed	Number of cases/controls	Type of control	Data collection	Matching factors	Adjustment factors ^{a)}
Segi <i>et al.</i> (1957)	Nationwide	1948–1952	644/3878 ^{b)}	Visitors at public health centers	Not reported	None	Age
Wynder <i>et al.</i> (1959)	Tokyo	Not reported	116/116	Hospital controls	Interview	Age	None
Yuasa and MacMahon (1970)	Tokyo	1965–1967	861/2268	Hospital controls	Interview	Position of bed	Age at first pregnancy
Hirohata <i>et al.</i> (1985)	Fukuoka	Not reported	212/424	Neighborhood controls (n=212) Hospital controls (n=212)	Interview	Age Living area	Age at first birth Parity
Kato <i>et al.</i> (1992)	Tokyo and other 7 prefectures	1990–1991	908/908	Hospital controls (n=664) Screening controls (n=244)	Self-administered questionnaire	Age	None
Wakai <i>et al.</i> (1995)	Tokyo	1990–1991	300/900	Hospital controls	Self-administered questionnaire	Age	Age at menarche Age at first birth Parity Age at menopause
Hirose <i>et al.</i> (1995)	Aichi	1988–1992	1052/21,299	Hospital controls	Self-administered questionnaire	None	Age
Hu <i>et al.</i> (1995)	Gifu	1989–1993	145/338	Neighborhood controls (screening participants)	Self-administered questionnaire	Age Living area	Age at menarche Age at first birth Parity Age at menopause

a) The variables for menstrual and reproductive events or age for mutual adjustments.

b) Number of original controls.

borhood controls. As the exposure variables of interest to us refer to notable events in life, the information is accessible and there is little opportunity for misclassification bias.

The results of the study reanalyses are summarized in Table II. Many of the OR estimates in the original reports were recalculated as necessary by the methods described earlier.

The homogeneity assumption was not violated for any exposure variable. The combined OR was significantly lower for women with onset of menstruation after age 16

compared to those before age 14 (Table III). Significantly higher OR was observed for women in any age group for first birth after age 25 compared to women with first birth before age 25. Nulliparous women were also at elevated risk compared to women with first birth before age 25, but the risk increase was significantly lower than for women with first birth after age 35. Although premenopausal women had a significantly higher risk than women with menopause before age 50, increased OR was not noted for women with menopause after age 50.

Table II. Summary of Reanalysis of Eight Studies

Study	Age at menarche		Age at first birth		Parity		Age at menopause	
		OR ^{a)} SE ^{b)}		OR SE				OR SE
Segi <i>et al.</i>	-13	1.00	-24	1.00				
	14-15	0.82 (0.19)	25-29	1.51 (0.16)				
	16+	0.50 (0.21)	30-34	1.96 (0.28)				
			35+	3.56 (0.36)				
Wynder <i>et al.</i>	-13	1.00	Null ^{c)}	—	1	1.00		
	14-15	1.14 (0.44)			2	0.72 (0.53)		
	16+	0.76 (0.45)			3+	0.51 (0.42)		
Yuasa and MacMahon	-13	1.00			1	1.00	-49	1.00
	14-15	0.95 (0.20)			2	1.16 (0.26)	50+	1.27 (0.27)
	16+	0.62 (0.18)			3+	0.69 (0.25)	Pre ^{d)}	—
Hirohata <i>et al.</i>	-13	1.00	-24	1.00	1	1.00		—
	14-15	0.82 (0.27)	25-29	1.58 (0.55)	2	1.74 (0.34)		
	16+	0.56 (0.24)	30-34	1.30 (0.67)	3+	1.26 (0.34)		
			35+	3.54 (1.06)				
			Null	3.56 (0.60)				
Kato <i>et al.</i>	-13	1.00	-24	1.00	1	1.00	-49	1.00
	14-15	1.15 (0.18)	25-29	1.25 (0.15)	2	0.83 (0.28)	50+	0.86 (0.26)
	16+	0.90 (0.18)	30-34	1.55 (0.23)	3+	0.71 (0.28)	Pre	1.99 (0.27)
			35+	1.93 (0.19)				
			Null	1.85 (0.19)				
Wakai <i>et al.</i>	-13	1.00	-24	1.00	1	1.00	-49	1.00
	14-15	1.00 (0.22)	25-29	1.03 (0.22)	2	0.92 (0.31)	50+	0.79 (0.26)
	16+	0.96 (0.26)	30-34	1.64 (0.30)	3+	0.72 (0.35)	Pre	2.22 (0.27)
			35+	2.17 (0.44)				
			Null	1.55 (0.25)				
Hirose <i>et al.</i>	-13	1.00	-24	1.00	1	1.00		
	14-15	0.99 (0.16)	25-29	1.33 (0.14)	2	0.91 (0.26)		
	16+	0.79 (0.17)	30-34	1.73 (0.14)	3+	0.72 (0.27)		
			35+	2.28 (0.14)				
			Null	1.31 (0.15)				
Hu <i>et al.</i>	-13	1.00	-24	1.00	1	1.00	-49	1.00
	14-15	0.88 (0.27)	25-29	1.40 (0.25)	2	0.36 (0.45)	50+	1.12 (0.39)
	16+	0.57 (0.31)	30-34	2.18 (0.55)	3+	0.18 (0.48)	Pre	3.09 (0.48)
			35+	3.43 (1.00)				
			Null	2.36 (0.73)				

a) Odds ratio estimates are adjusted for the other listed variables.

b) Standard error of the natural logarithm of the OR.

c) Nulliparous.

d) Premenopausal.

Table III. Combined Odds Ratio (OR) Estimates of Menstrual and Reproductive Factors for Breast Cancer among Japanese Women

Variable	OR	95% CI ^{a)}
Age at menarche		
-13	1.00	
14-15	0.96	(0.83-1.12)
16+	0.68	(0.59-0.77)
Age at first birth		
-24	1.00	
25-29	1.32	(1.14-1.53)
30-34	1.71	(1.41-2.09)
35+	2.26	(1.85-2.77)
Nulliparous	1.56	(1.27-1.91)
Parity		
1	1.00	
2	0.95	(0.75-1.20)
3+	0.68	(0.54-0.86)
Age at menopause		
-49	1.00	
50+	0.98	(0.74-1.31)
Premenopausal	2.21	(1.53-3.20)

a) 95% confidence interval.

Although individual ORs for women with 3 or more births compared to uniparous women ranged from 0.18 to 1.26 across the studies, homogeneity was not rejected; chi-square=11.53 on df=7. A significant protective effect of high parity was noted even after adjusting for age at first birth and other menstrual factors.

DISCUSSION

Our analyses confirmed that late age at first birth, early age at menarche, and premenopausal status were significantly associated with risk of breast cancer among Japanese women. We also found that parity is one of the independent risk factors of breast cancer.

MacMahon *et al.*²⁹⁾ suggested that lower risk associated with high parity was explained by the association of high parity and low age at first birth. Since then, however, the independent protective effect of high parity has been confirmed in many studies. Ewertz *et al.*⁵⁾ conducted a meta-analysis of 18 studies from the Nordic countries and estimated a risk reduction of about 16% for every 2 births.

Combined effect, or average effect, can be representative of the component studies under the homogeneity assumption. Somewhat conflicting results were observed for the effect of high parity across the studies. The characteristics of the population examined in the study might be related to the results obtained. Five hospital-based case-control studies showed similar values of ORs for each exposure variable. The results of the two studies

using neighborhood controls were conflicting. The reason why the magnitudes of ORs for parity were the largest in the study by Hirohata *et al.*¹¹⁾ and the smallest in the study by Hu *et al.* is unclear. Although we cannot rule out the possibility that both or either of them are affected by bias, it might be due to the small population numbers of these studies with neighborhood controls.

The level of exposure in the study population is different across the studies. The distribution of number of births was skewed to high parity among the populations of the early studies. Therefore, the actual level of the highest category for parity must be higher in the early studies than the new ones. In fact, the protective effect of 3 or more births was stronger in the studies using hospital controls by Wynder *et al.*⁹⁾ and Yuasa and McMahon¹⁰⁾ than in the recent three hospital-based case-control studies.

To examine the influences of these study-by-study characteristics, regression meta-analysis using a random effect model which includes such characteristics as covariates is useful.³⁰⁾ Although we could not apply these methods because we have at most eight component studies, the results of the homogeneity test indicate that the study variation is small enough to be reasonably ignored in the present meta-analysis.

Several limitations of this study should be considered. An advantage of meta-analysis is that it allows examination of subgroups when individual studies have inadequate numbers, but we could not take full advantage of this. Taking account of the generally accepted hypothesis that the effects of menstrual and reproductive events vary with time since occurrence,³¹⁾ age should be considered as a potential effect modifier. Such an interacting effect may not have introduced variation of study results because age distributions were apparently not greatly different across the studies. However, examining the effects of exposures according to age at risk would be valuable to explore the observed changes in the pattern of breast cancer incidence among Japanese women. We could not conduct separate analyses according to age at risk because of insufficient information from published data.

The menstrual and reproductive events might also affect breast cancer risk differently in pre- and postmenopausal women. We performed separate combination of ORs for pre- and postmenopausal groups based on only 4 studies in which we could obtain the necessary information. The summary ORs for two categories for parity (number of births of 2 and 3 or more compared to uniparity) were 0.74 (95% confidence interval (CI): 0.49-1.13) and 0.61 (95% CI: 0.38-0.98), respectively, for premenopausal women, and 0.94 (95% CI: 0.55-1.60) and 0.84 (95% CI: 0.48-1.46), respectively, for postmenopausal women. High parity was more strongly associated with risk in premenopausal women, although

the difference between pre- and postmenopausal women was not statistically significant.

The process of extracting the summary estimates from individual studies may have introduced bias. We applied the methods for adjusting for confounders or converting the exposure categories because we could not obtain complete information from the original publications. However, taking account of the sizes of ORs and their variances across the studies, inaccuracies introduced from this process are expected to be minor and unlikely to be of great consequence.

There is a concern that excessive credibility might have been assigned to poor-quality studies by summarizing them in the quantitative analyses. Setting standards for minimal acceptable quality in undertaking a meta-analysis is recommended.³²⁾ Within the context of meta-analysis, stratification of studies on quality scores is also useful to clarify the effect of the study quality on the results.^{33,34)} We assessed the quality of each study to determine whether it should be included in the present meta-analysis. However, because of scant study descriptions, we did not carry out analysis based on scoring of the component studies. It should be noted that in effect, larger studies, mainly hospital-based case-control studies

in the present meta-analysis, are given more importance in summarizing measures than smaller studies in spite of the possible unevenness of study quality. We recalculated the combined ORs after dropping the largest study,¹⁴⁾ but the results were not substantially altered.

We assumed that the magnitudes of the effects of these risk factors had not changed over time and included studies covering a relatively long period. Although the null hypotheses of homogeneity were not rejected, it is possible that the real effect sizes might have changed during the period. The small number of studies precluded us from analyzing the influence of the study period using the stratification or regression technique.

The present meta-analysis provided substantial improvements in the precision of effect sizes. Although the individual studies used different methods and populations, we verified that the effect sizes are relatively homogeneous. This encourages us to take further advantage of the methods of meta-analysis to examine the variation in risk estimates for menstrual and reproductive factors by age at risk, menopausal status and other population characteristics. Joint effects of 3 variables for menstrual and reproductive factors are also to be evaluated.

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